



UNIVERSITY OF ENGINEERING AND TECHNOLOGY MARDAN

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REG NO	23MDBCS435
SECTION	A
DEPPT	COMPUTER SCIENCE
SUBJECT	DLD LAB
TEACHER	SHAHZAD SIR

LAB TASK : 8

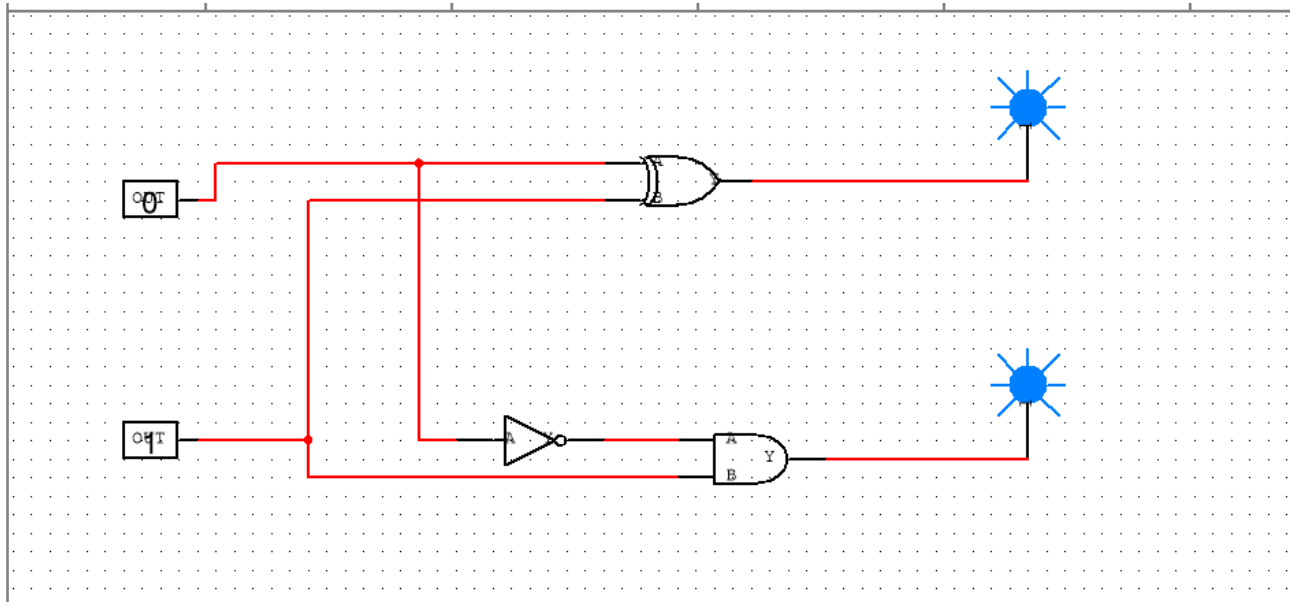
Half Subtractor

Logic functions:

$$D_i = A + B$$

$$B_o = A \cdot B$$

CIRCUIT DIAGRAM:



Input Output

Input		Output	
A	B	Bo	Di
1	1	0	0

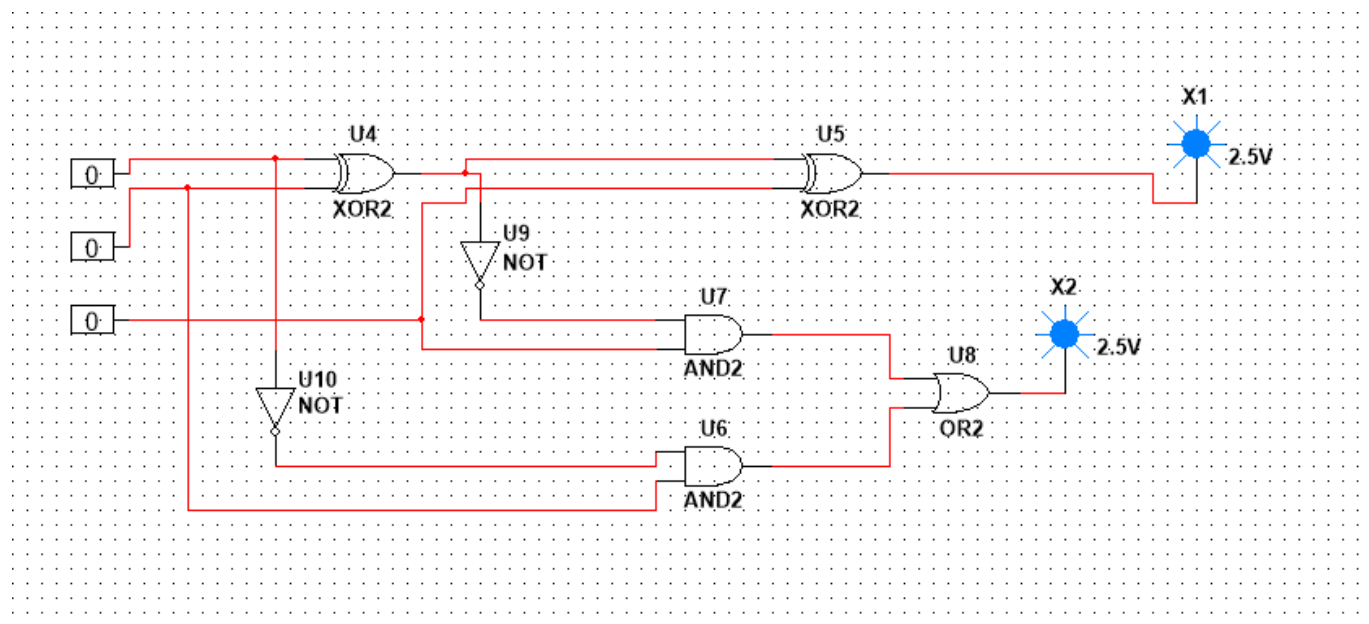
Full Subtractor

Logic functions:

$$D_i = A + B + B_{in}$$

$$B_{out} = (B_{in} \cdot (A + B)) + (A \cdot B)$$

CIRCUIT DIAGRAM:



TRUTH TABLE:

A	B	Cin	Bout	Di
0	0	0	0	0
0	1	0	1	1
1	0	0	0	1
1	1	0	0	0

THE END